Rahul Rustagi

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Education

Georgia Institute of Technology — Atlanta, GA
MS in Electrical and Computer Engineering | Depth: System and Controls — GPA: 4.0/4.0
Coursework: Multi-Agent Control, Computer Vision, PDEs in Image Processing — Aug. 2024 – May 2026

Indian Institute of Technology — Kanpur, India
BTech in Aerospace Engineering | Depth: Aerial Robotics — GPA: 9.2/10
Minors in: Machine Learning, Computer Systems, English Literature — Aug. 2020 – May 2024

Technical Skills

Robotics: ROS 1/2, Gazebo, OpenCV, Isaac Gym, Moveit, RViZ, PX4 | **Frameworks**: Detectron2, TensorFlow, PyTorch
Programming: C/C++, Python, OpenMPI, CUDA, MATLAB | **Tools**: Git, Bash, Docker, CMake, PyCharm, Qt5

Publications

1. **Vision-Based** Autonomous Ship Deck landing of an Unmanned Aerial Vehicle using Fractal ArUco markers
   C. Prachand, R. Rustagi, R. Shankar, J. Singh, A. Abhishek, K.S. Venkatesh | 2025 AIAA SciTech Forum
2. Lifetime Improvement in Rechargeable Mobile IoT Networks Using **Deep Reinforcement Learning**
   A. Singh, R. Rustagi, R. M. Hegde | 2023 IEEE Transactions on Circuits and Systems II: Express Briefs
3. Mobile Energy Transmitter Scheduling in Energy Harvesting IoT Networks using Deep Reinforcement Learning
   A. Singh, R. Rustagi, R. M. Hegde | 2022 IEEE 8^{th} World Forum on Internet of Things Conference

Work Experience

Intelligent Vision and Automation Laboratory, Georgia Tech — Aug 2024 – Ongoing
Graduate Research Assistant | Area: Robot Perception and Navigation | Guide: Dr. Patricio Vela — Atlanta, Georgia
- Implemented a **Visual SLAM system** for **Autonomous Navigation** in simulated **GPS-denied environments**
- Implemented Direct Sparse Odometry Lite (DSOL) for direct image alignment and camera pose estimation
- Understood **Scene Reconstruction**, **Photometric Bundle Adjustment**, Image Pyramid to improve pose tracking
- Refined pose estimation feedback by fusing IMU data using **Factor Graph** for smoothness in a closed-loop system

Helicopter and VTOL Laboratory, IIT Kanpur — Jan 2024 – Jul 2024
Project Associate | Area: Vision-Guided Control with Deep Learning | Guide: Dr. Abhishek — Kanpur, India
- Designed a pipeline[1] to predict landing platform 6 DoF motion, leveraging **deep learning** for UAV trajectory planning
- Employed Fractal Fiducial markers for estimating platform 6DoF pose at 60Hz using a Realsense D435i camera
- Deployed a **custom LSTM model** on **Jetson Nano**, integrating **TF-TRT & quantization** for real-time performance
- Applied **QP Solver** calculating future 2 seconds of optimal time-constrained trajectory waypoints for a safe touchdown

Advanced Robotics Optimization and Control Laboratory, Carleton University — May 2023 – Dec 2024
Research Intern | Area: Navigation and Vision-Based Guidance | Guide: Dr. Chao Shen — Ottawa, ON
- Improved robot's position estimates in a **non-static environment** through **bayesian sensor fusion** using image data
- Conducted ArUco marker detection and pnp pose estimation to embed vision data with the map of the environment
- Built an algorithm to use vision pose estimates to filter out unstable **AMCL** estimates in moving environment
- Calculated **ATE** and **RTE** accuracy metrics using **evo** for mentioned localisation algorithms with max error $\leq$ **1m**

Wireless Sensor Network and IoT Laboratory, IIT Kanpur — May 2022 – Dec 2022
Research Assistant | Area: Reinforcement Learning | Guide: Dr. Rajesh Hegde — Kanpur, India
- Employed Reinforcement Learning algorithm to **autonomously decide** charging order in low-powered **IoT** environment
- Constructed a RL environment using **OpenAI-gym** and simulated a network of 10 IoT nodes using **pybullet** physics
- Trained **TD3-PG, DDPG** and **PPO** algorithms with TD3-PG converging **20%** faster[3] to optimal and higher reward
- Deployed the algorithm on a mobile IoT network with random-walk & presented novel improvement[2] to mobile systems

Selected Projects

Advanced Computer Vision Projects | Course Project CS6476 @ GT
Skills Acquired: ViT, 3D Reconstruction, Visual Odometry, PyTorch, OpenCV Aug 2024 – Dec 2024

- Built a custom **Vision Transformer** and trained it on a fraction of **CIFAR-10** dataset to understand accuracy metrics
- Implemented **R-CNN**, **HoG**, and **YOLO** models in pytorch and compared accuracies against **Microsoft-CoCo** dataset
- Benchmarked **PSPNet** on **Camvid-11** dataset against other models like **ResNet-50** and a custom **SegmentationNet**
- Performed **Structure from Motion** to determine **visual odometry** using **epipolar geometry** and **image warping**

MAV Swarm Formation Challenge | Drona Aviation github/interiit11
Skills Acquired: C++, Embedded Programming, OpenCV, Multi-Threading, Robotics Jan 2023 – Mar 2023

- Developed a **ROS-independent** control pipeline in C++ for visual feedback-based square formation of 4-MAV swarm
- Designed a parallelized camera driver, increasing vision-based detection ArUco frequency from 27Hz to 55Hz
- Implemented **multi-threaded** position controllers for centralized control of 4 MAVs with real-time trajectory tracking
- Ensured **thread-safe synchronization** using **mutex-based deadlock prevention** for seamless MAV swarm operation

Multi-Payload Delivery Challenge using UAV | Flipkart GRID 4.0 github/shastra23
Skills Acquired: ROS, QGroundControl, PX4, OpenCV, Boost, Arduino IDE Nov 2022 – Jan 2023

- Developed an **end-to-end autonomous ROS pipeline** for UAV payload pickup & drop-off scattered in a 10x10m field
- Implemented grid-search through **QGc** and used **OpenCV** for 6DoF pose estimation of payloads with ArUco markers
- Programmed an **Arduino Due** to control an electromagnet actuator enabling autonomous payload pickup when detected
- Executed the pipeline on **Odroid XU4** Flight Computer with **PX4** stack communicating using **Mavlink** protocol

Position of Responsibilities | Leadership Experience

Software Head github/aerial
Aerial Robotics, IIT Kanpur | Team Advisor: Dr. Twinkle Tripathy May 2022 – Jul 2023

- Led a contingent of 5 members to participate in national level competitions in robotics representing the institute
- Successfully bagged podium finish at the **InterIIT Tech Meet 11.0** and **10.0** to win Silver and Bronze Medal
- Responsible for developing and maintaining the software stack for the team's custom-built fleet of aerial robots
- Conducted workshop on ROS and OpenCV for **200 students** using an interactive programming assignment

Achievements

Awards: Academic Excellence, 2^{nd} Best Outgoing Student, Silver at InterIIT Tech Meet 10, Bronze at InterIIT Tech Meet 11
Grants: MITACS GRI 2023, MCM (at IIT Kanpur), Under 50 All India Rank in CMI Institute, INSPIRE Scholarship

Coursework

Machine Learning: Introduction to Machine Learning, Probabilistic Machine Learning, Introduction to Reinforcement Learning, Neural Networks and Deep Learning *(Online)*
Robotics : Computer Vision, PDEs in Image Processing, Intro to Robotics Research, Multi-Robot Systems
Controls : Multi-Agent Controls, Aircraft Control Systems, Optimal Space Flight Control
Systems : Embedded and Cyber Physical Systems, Software Development and Operations, Data Structures and Algorithms, Computer Networks